Distance and midpoint

1 Find the distance between the following pairs of points. Round your answers to one decimal place where necessary.

a $(-2, 4, 9)$ and $(5, 0, -1)$

b $(1, 3, -2)$ and $(4, -2, 0)$

c $(-1, -2, 4)$ and $(-6, 1, 3)$

d $(0, 6, 10)$ and $(-5, -2, -1)$

2 Find the midpoint between the following pairs of points:

a $(-5, 0, 6)$ and $(4, 3, -2)$

b $(4, 3, -5)$ and $(3, -7, 1)$

c $(-6, 8, 0)$ and $(-4, 6, 1)$

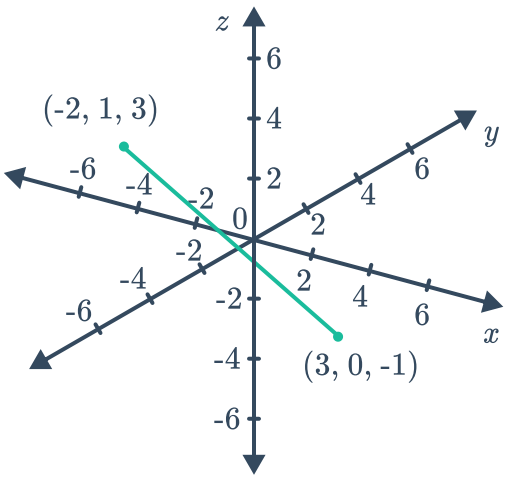
d $(21, 20, 11)$ and $(7, -3, 9)$

3 For each of the following line segments, find:

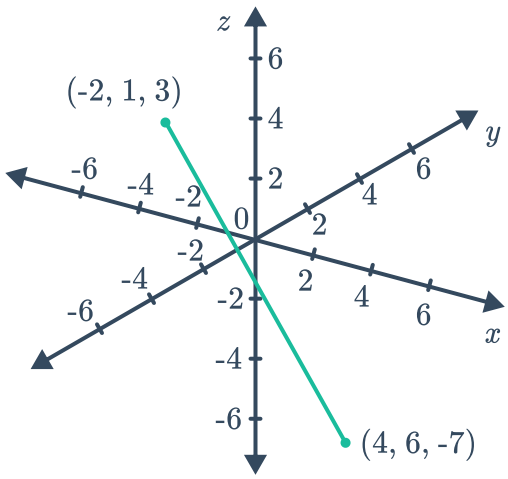
i The length to two decimal places.

ii The midpoint.

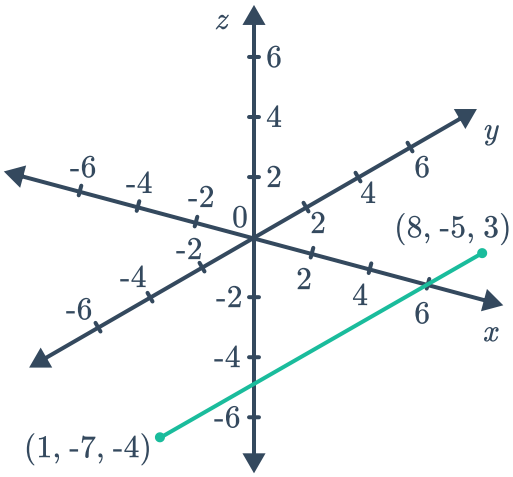
a



b



c



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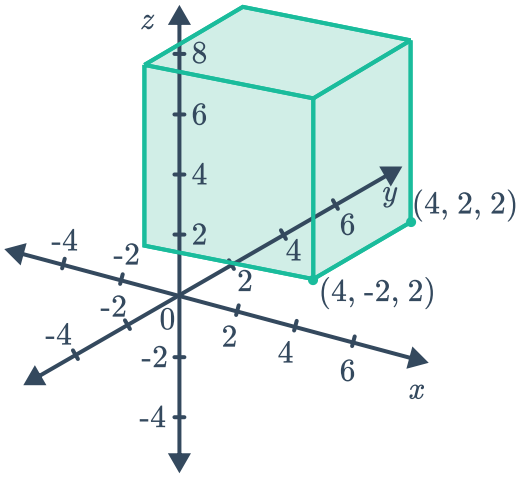
Applications

4 Consider the cube shown:

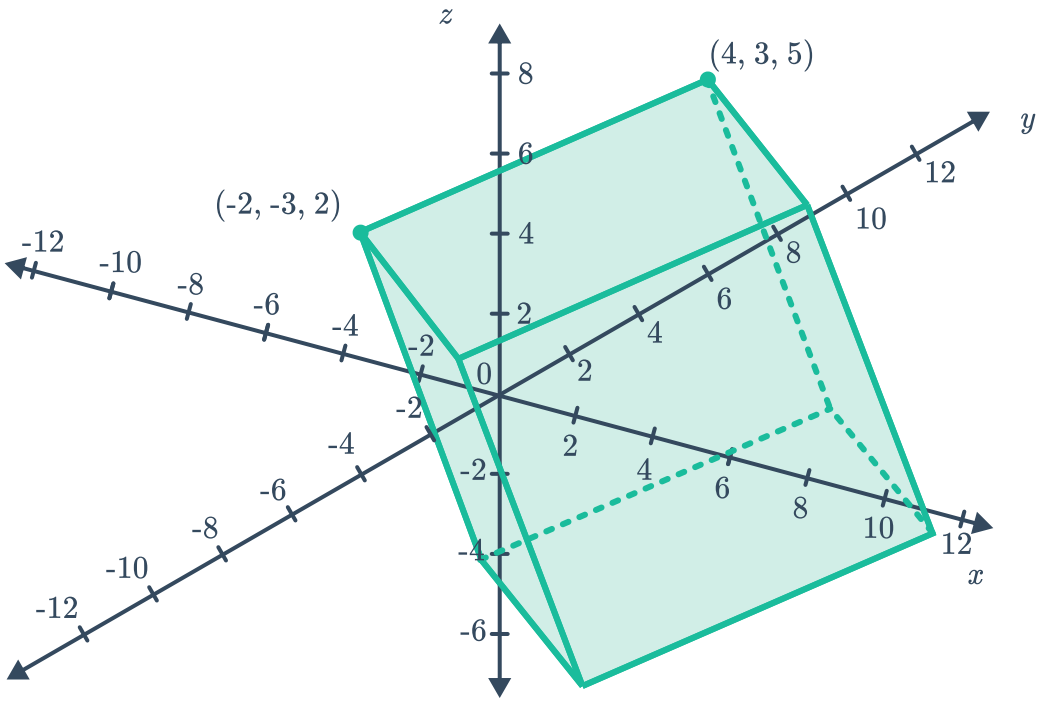
a Find the side length of the cube.

b Find the surface area of the cube.

c Find the volume of the cube.

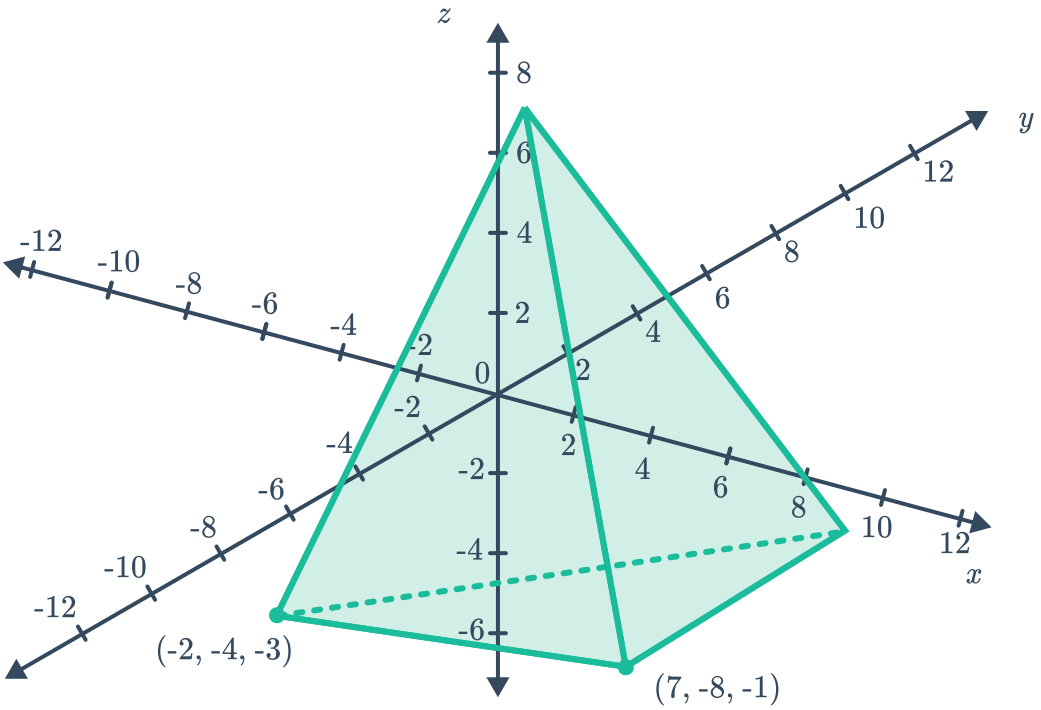


5 Find the volume of the cube shown:



6 Find the surface area of the following regular tetrahedron. Round your answer to one decimal place.

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